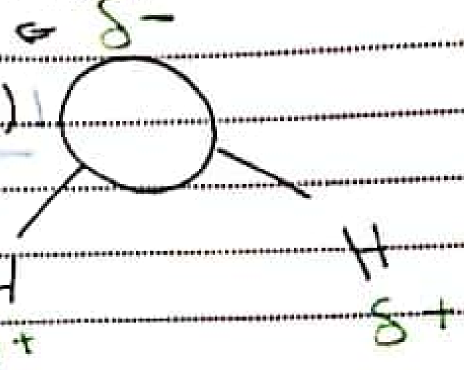
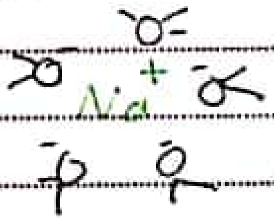
Subject: Stoichiometry (II)

→ Electrolyte solution [NaCl]
→ Non electrolyte solution [$C_6H_{12}O_6$]

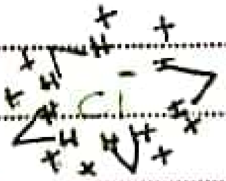
attraction (+/-)
repulsion (+/+ | -/-) = ($2H^+$)

cuz all is (0)



all direction


⇒ weak bond & attraction and repulsion at the same time

all directions


molecule ⇒ Ions (✓) ⟷ $NaCl \rightarrow Na^+ + Cl^-$
molecule ⇒ molecule (✗) ⟷ $C_6H_{12}O_{6\,(s)} \rightarrow C_6H_{12}O_{6\,(aq)}$

still found
$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$ ⇒ weak < 100%

$HCl \rightarrow H^+ + Cl^-$ ⇒ Strong = 100% dissociation

Solubility :- amount of substance dissolved in given quantity of solvent at a certain Temperature

* Solute :- مذاب ie NaCl ⇒ less amount
* Solvent :- مذيب ie H_2O ⇒ more amount
* Solution :- (NaCl / H_2O)

Subject : Neutralisation reaction

→ $HCl + NaOH \rightarrow NaCl + H_2O$

polar + polar

→ Both of them are limiting reactant
→ if one of them is still there
The whole solution is the same

* 1A are strong Bases also → 2A

oxidation Number :

(H) → alone (+1) HCl
(H) → metals (−1) NaH

(O) → (−2) usually
(O) → H_2O_2 | OF_2 (+2)

* $HCl + Na_2S \rightarrow NaCl + H_2S$

* $HCl + NaHCO_3 \rightarrow NaCl + H_2CO_3$

يتفكك

CO_2 H_2O (Biol)

Subject: exchange reaction

$AgNO_3 + KCl \rightarrow \underset{(s)}{AgCl} + KNO_3$

oxidation - reduction reactions

$\underset{(s)}{Zn} \rightarrow Zn^{+2} + \cancel{2}\bar{e}$ losing ~~lost~~ electrons

$Cu^{+2} + \cancel{2}\bar{e} \rightarrow \underset{(s)}{Cu}$ gaining electrons

reduction

$\underset{(s)}{Zn} + \underset{(aq)}{Cu^{+2}} \rightarrow \underset{(aq)}{Zn^{+2}} + \underset{(s)}{Cu}$

oxidation

* Ions always in water

activity series

ease of reduction increases

$Zn \rightarrow Zn^{+2} + 2\bar{e}$

$H_2 \rightarrow 2H^{+} + 2\bar{e}$

$Cu \rightarrow Cu^{+2} + 2\bar{e}$

ease of oxidation increases

* $H_2 \rightarrow 2H^{+} + 2\bar{e}$

$Cu^{+2} + \cancel{2}\bar{e} \rightarrow Cu$

$\underset{g}{H_2} + \underset{(aq)}{Cu^{+2}} \rightarrow \underset{(aq)}{2H^{+}} + \underset{(s)}{Cu}$

Subject: Molarity

$$M = \frac{n_{solute}}{V_{solution}} = \frac{mol}{L} = mol/L \ (M)$$

Solute => Mol

Ex When 18 g $C_6H_{12}O_6$ is desolved in 200 ml solution what's the molarity of the solution

$$n = \frac{m}{m.m} = \frac{18}{180} = .1$$

.2 L

C : 12
O : 16
H : 1

$$= \frac{.1}{.20} = .5 \text{ Molar}$$

Ex $2HCl + Mg(OH)_2 \rightarrow MgCl_2 + 2H_2O$

18.25 g — 100 ml — ? g

.5 molar

$M = \frac{n}{L}$

Mg : 24
Cl : 35.5
O : 16
H : 1

$HCl = 1 + 35.5 = 36.5$

$$n = \frac{18.25}{36.5} = 0.5$$

$$0.5 = \frac{n}{.1} = 0.05$$

$MgCl_2$

$\boxed{95} = 24 + (35.5 \times 2)$

$Mg(OH)_2$ 1 → 1 $MgCl_2$

0.05 → X

$$X = 0.05 \text{ mol} = \frac{m}{M.m} \Rightarrow m = 4.75$$

Subject: Dilution

$M = \frac{w}{V}$ (w → Constant)

20 mL H_2O

100 mL 0.1 M HCl

$$M_1 V_1 = M_2 V_2$$

Before adding: After adding: [the whole solution]

$0.1 \times 100 = M_2 \times [120]$ 100 + 20

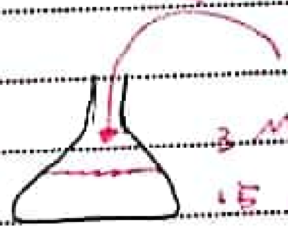
$M_2 = 0.083$ molar

[illegible] ⇒ Dilution

Q

3 M HNO_3 ⟹ 450 mL 0.1 M

435 mL H_2O



3 M 15 mL

$M_1 V_1 = M_2 V_2$

$3 \times V_1 = 0.1 \times 450$

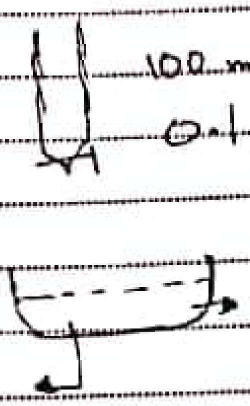
$\boxed{V_1 = 15 \text{ mL}}$

$M_1 V_1 = M_2 V_2$

$3 \times 15 = M_2 \times 450$

$M_2 = 0.1$

Subject: Titration



$HCl + NaOH \rightarrow NaCl + H_2O$

30 mL   1 M

0.1 M   ?

→ reacting volume

Ex) How many grams of $Ca(OH)_2$ are needed to natrlize 250 mL of .3 M HCl.

Ca : 40
O : 16
H : 1

$Ca(OH)_2 + 2HCl \rightarrow CaCl_2 + 2H_2O$

?g   250 mL

.3 M

$n = 0.075$ mole HCl

$2 \rightarrow 1$

$0.075 \rightarrow X$

$X = \square = \frac{m}{Mm}$ ...

Subject: Electromagnetic

λ → ↑

E → ↓

Gamma rays — X-rays — UV — vis — I.R — Microwave — Radio Frequency

V I B G Y O R

λ → small

E → ↑

λ → big

E → ↓

$E = \boxed{h}\,\nu$ → Planck's constant

proportional

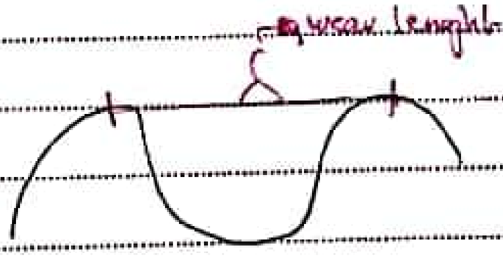
constant

$E = \boxed{h}\,\dfrac{\boxed{c}}{\lambda}$

inversely proportional

$c = 3 \times 10^{8}$ m/s

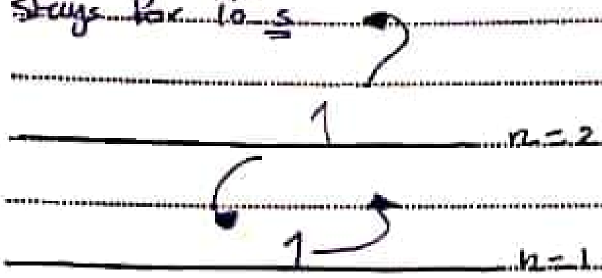
$E = h\,\dfrac{c}{\lambda} \quad \dfrac{m/s}{m} = \dfrac{1}{s} = H(s^{-1})$



* $h = 6.63 \times 10^{-34}$ J.S

continued

stays for 10^{-14} s



$$\Delta E = 2.18 \times 10^{-18} \left(\frac{1}{(n_1^2)} - \frac{1}{(n_2^2)} \right)$$

Jol Jol

* λ even appotual.

ΔE → (−) ↓ emission
ΔE → (+) ↑ absorbition

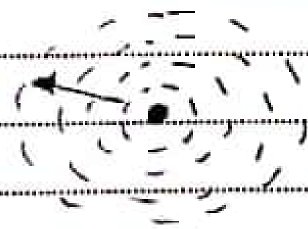
from where to where

Ex) Which of the following transitions has the highest wave length :-

→ $n_1 \to n_2$ → $n_2 \to n_3$ → $n_3 \to n_4$

Ans :- highest λ = lowest energy so

low energy



light

wave	partical
γ	m
λ	v
h	m.v ⇒ momentum

only in waves

* light has a mixed proprtion of a partical and a wave

Subject:

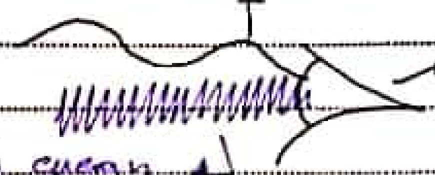
De Broglie

$J = Kg \frac{m^2}{s^2}$

$\lambda = \frac{h}{mv}$ h → constant

inversally

$m = \frac{Kg \frac{m^2}{s^2} \times s}{Kg \times \frac{m}{s}} = \boxed{m}$



$\boxed{Ex}$ The λ of an electron (in nanometer) to move with a velocity of 500.000 m/s is $m_e = 9.11 \times 10^{-31}$ Kg

A → $\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-9}}{9.11 \times 10^{-31} \times 5 \times 10^{5}} =$

Subject : Quantum Numbers / /

De Broglie

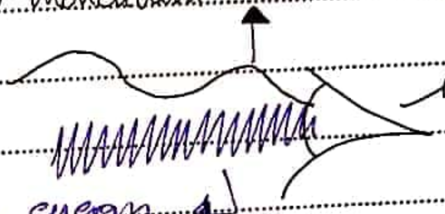
$J = Kg \frac{m^2}{s^2}$

$\lambda = \frac{h}{mv}$ $h \rightarrow$ constant

$m = \frac{Kg \frac{m^2}{s^2} \times s}{Kg \times \frac{m}{s}} = \boxed{m}$

reversally

low momentum



high energy

$\boxed{Ex}$ The λ of an electron (in nanometer) to move with a velocity of 500,000 m/s is $m_e = 9.11 \times 10^{-31}$ Kg

A→ $\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-9}}{9.11 \times 10^{-31} \times 5 \times 10^{5}} =$

→ Uncertinaty princible :-

momentary

$\Delta x \cdot \Delta mv \geq \frac{h}{4\pi}$

explination :- if you can find Δx then you are so far away from finding (Δmv) The Answer will be uncertain

Subject: Quantum numbers

* Principle ~~number~~ Quantum # (n)
* Angular " " " (l)
* Magnatic " " " (ml)
* Spin " " " (ms)

$Na: 1s^2 2s^2 \underbrace{2p^6}\, 3s^1$
11

3de electron in 2P

$n = 2$ $\quad l = 1$

$l = 1$ $\quad m_s = +1/2$

↑↓ ↑↓ $\boxed{↑}$↓

"زي ما عبيت بتعد"

محمد فارس

* ms ↑ → +
↓ → −

H2O_GROUP

$\underset{26}{Fe}: 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 \underbrace{3d^6}$

$n = 3$ $\quad l = 2$
$ml = +1$ $\quad m_s = +1/2$

4th electron in 3d

↑↓	↑	↑	$\boxed{↑}$	↑
2−	1−	0	1	2

$\underset{54}{Xe}: 5s^2 4d^{10} \underbrace{5p^6}$

$n = 5$ $\quad l = 1$
$ml = -1$ $\quad m_s = -1/2$

↑$\boxed{↓}$	↑↓	↑↓
1−	0	+1

N O T E B O O K

Subject : .... / /

* Pauli exclution : No electron can have the same quantum numbers in the same Atom.

→ it may have the same $\underline{3}$ numbers except (m_s)

← معنى يتشابه بكل الأعداد ماعدا m_s

$\underset{24}{Cr}$: $[\underset{18}{Ar}]$ $\underline{4s^2}$ $3d^4$ → half filled

(half filled ↑ $4s^2$)

$[Ar_{18}]$ $4s^1$ $3d^5$ ⇒ more stable
half filled.

Only between $\underline{(s,d)}$

Periodic Table.

[1] effective nuclear charg (Z_{eff}) قوة سحب الالكترونات

← كلما كانت النواة قويه بتسحب الالكترونات بقوه اكبر بتقل مرحلة [illegible]

$$Z_{eff} = \underset{\text{protons}}{Z} - \underset{\text{shield electrons}}{S}$$

$\underset{3}{Li}$: $1s^2\ 2s^1$ $Z_{eff} = Z - S$
$= 3 - 2 = \boxed{1}$

Subject :

Be : $1s^2\ \boxed{2}s^2\ \boxed{2}p^1$
5

$Z_{eff} = Z - S$
$5 - 2 = \boxed{3}$

shield e
بعتبر shield
بنفس المدار

في الدورة الواحدة Z eff ثابتة
مثلاً الدورة الأولى Z eff = $\boxed{1}$

في الدورة الواحدة تزيد Z eff كلما اتجهنا من اليسار لليمين

Subject: Z.eff / Ionization Energy / /

Be : $1s^2\,\boxed{2}s^2\,\boxed{2}p^1$ — Z.eff = Z − S

5 — 5 − 2 = $\boxed{3}$

shield e — يتم

بنفس المدار — لا

shield يعتبر ×

* في المجموعة الواحدة Z.eff ثابتة

مثلاً المجموعة الأولى Z.eff = $\boxed{1}$

في الدورة الواحدة تزيد Z.eff كلما تحركنا من اليسار لليمين

Ionization Energy : Amount of energy required to remove Valance electrons to ∞ [in gaseous state]

$Na_{(g)} \longrightarrow Na^+ + \bar{e}$ $I_1 = +500$ KJ/mol

endo مطلوب (+)

exo مطلق (−)

$_{11}Na : 1s^2\,2s^2\,\underline{2p^6}\,\underline{3s^1}$

2nd 1st

I.e increases + Z.eff

I.e dec slightly

← يلزم طاقة أقل مما كانا للأسفل

أقل.

Z.eff = P − Z

stable difference

$_{11}Na : 1s^2\,2s^2\,2p^6\,3s^1$

① Z.eff = 11 − 10 = 1

② Z.eff = 11 − 2 = 9

much more

Subject: I.e / size of Ions

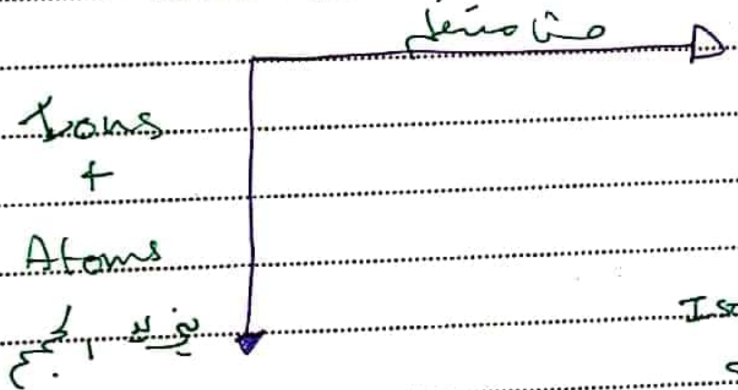
the first time from stable → Ion → has a (−)
So it's difficult because of the attraction

1- e / +   *(Na) has no 3rd I.e
2- الغريب من ، الشاذه
3- Z.eff

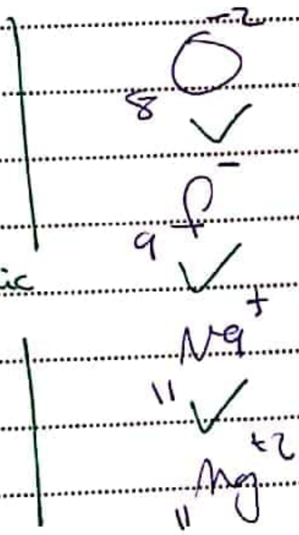
Mg : $1s^2 2s^2 2p^6 3s^2$
12

① 12 − 10 = 2   700 ⟶ attraction
② 12 − 10 = 2   1500
③ 12 − 2 = 10   7700

Size of Ions



Iso electronic
seirs
Same # electrons



As charg increases
size decreases

Rate the following Ions in order of decreasing size:-
Mg / $_{12}mg^{+2}$ / ca^{+2}

$Mg > mg^{+2} > ca^{+2}$

Subject : Electron affinity / /

exeptions in I.e.

Li Be B C N O F

$I_{Be} > I_B$ $I_N > I_O$

more stable ← more stable ←

def: Energy for the Introduction of an electron to the Velance Shell in the gases State.

$Cl_{(g)} + \bar{e} \rightarrow Cl^{-}$ $\Delta E = +$ → almost

ه ماحقي طاقه بالسالب هي مجرد تغير عند طارد أو ماص

$-20 < -500$ compare the numbers not (charge)

ه أول ما يدخل الغلاف يطرد الطاقة كلها و سيصير عنده بتش طاقة الغلاف

→ increases

شروط

① لازم تحمل (V/E) ما يعاني الالكترونات المتكافئة؟

② (e) سالبه و أكبر بده

يدخل (−) ف repulsion

enough energy

ه — ← بحبه

+ ← بتكرهه

Subject :

Be / N > O (+) سالبة
O / F < O (-) موجبة

فوائد

* metal oxide + water → metal hydroxide *

$Na_2O + H_2O \rightarrow NaOH$
strong Base.

* metal oxide + acid → salt + H_2O
$Na_2O + HCl \rightarrow NaCl + H_2O$

* Nonmetal oxide + H_2O → acidy
$CO_2 + H_2O \rightarrow H_2CO_3$

* None metal oxide + Base → Salt + H_2O
$CO_2 + NaOH \rightarrow Na_2CO_3 + H_2O$

metalic charecter dec ↓

*